

Are Dollar Stores Magnets for Violent Crime? Evidence from Chicago

Urban Affairs Review
2024, Vol. 60(5) 1540–1576
© The Author(s) 2024
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/10780874241226875
journals.sagepub.com/home/uar



Eun Jin Shin 

Abstract

With the rise of economic inequality in recent decades, dollar stores have flourished in the United States. Although the media has increasingly portrayed dollar stores as violent crime magnets, limited academic research has examined their impact on nearby crime. This study employed a difference-in-differences approach to investigate the impact of dollar stores on localized violent crime patterns in Chicago. Results showed a significant increase in violent crimes, especially robberies, in nearby areas relative to the control areas, following the opening of dollar stores. There were no signs of spatial crime displacement. Moreover, after the closing of dollar stores, the levels of violent crime in nearby areas relative to the control areas returned to those of the prestore opening period, regardless of the type of crime. This study also explored how the impact of dollar store openings on nearby crime rates evolves over time and varies according to the surrounding context of the store's location.

Keywords

dollar stores, discount retailers, violent crime, Chicago

Department of Public Administration and Graduate School of Governance, Sungkyunkwan University, Seoul, Republic of Korea

Corresponding Author:

Eun Jin Shin, Department of Public Administration and Graduate School of Governance, Sungkyunkwan University, 25-2 Sungkyunkwan-ro, Jongno-gu, Seoul, Republic of Korea 03063.
Email: eunjinshin@skku.edu

Introduction

With the rise of economic inequality in recent decades, dollar stores experienced striking growth in the United States, especially after the Great Recession (Chenarides et al. 2021; Grigsby-Calage, Mullally and Volpe 2021). Along with the expansion of the dollar store industry, mass media has portrayed dollar stores not only as a threat to traditional food retailers selling healthy food (Brown 2018; Siegel 2019) but also as magnets for violent crimes generally involving armed robberies and sometimes, homicides (Jones 2019; MacGillis 2020). Consequently, residents' protests against opening dollar stores in their neighborhoods have become increasingly common in recent years, fueling policy debates (Brown 2018).

A small but growing body of literature has investigated how dollar stores affect local communities. While much of this research has concentrated on the influence of stores on the food retailing environment and food access (Drichoutis et al. 2015; Caspi et al. 2016), there has been less exploration into the broader impacts of dollar stores on local communities. Of the various potential effects of dollar stores on surrounding areas, this study focuses on the effect on violent crime.

There are several theoretically plausible reasons for a link between dollar stores and violent crime. According to routine activity theory (Cohen and Felson 1979), dollar stores may heighten violent crime, especially robbery, in nearby areas as their opening leads to a convergence of three elements—motivated offenders, suitable targets, and the absence of capable guardians—in space and time. These stores lack security guards and high-quality surveillance cameras that can serve as “capable guardians” against armed offenders (Meyersohn and Cohen 2020; Vargas 2021). Furthermore, dollar stores, typically found in socioeconomically disadvantaged neighborhoods, may increase the proximity of “potential offenders” to “suitable targets” (store employees and customers). However, dollar store customers could also serve as potential guardians by consciously or unconsciously monitoring the surrounding area, thereby reducing nearby crime; this aligns with the concept of “eyes on the street” developed in Jane Jacobs’s work (1961) on street life in urban neighborhoods.

Moreover, dollar stores are often considered a sign of neighborhood disorder due to their unattractive physical appearance (Kassab 2013; Shrestha 2016; Higgs 2020) and their association with poor ethnic neighborhoods (Shannon 2021). The broken windows theory posits that the presence of disorder stimulates fear of crime, weakening informal social control, which in turn results in crime (Wilson and Kelling 1982). While the relationship between neighborhood disorder and other types of crime is found to be weak (both theoretically and empirically), neighborhood disorder appears

to have a stronger, positive association with robbery rates (Sampson and Raudenbush 1999).

Literature on risky retailers and crime has further suggested a theoretical link between dollar stores and violent crime. From the perspective of routine activity theory, the customers of risky retailers (e.g., pawn shops, marijuana dispensaries) tend to carry cash for their transactions and, therefore, are more attractive targets for robberies (Kubrin et al. 2011); offenders targeting these types of retailers or their customers would prefer robbery over property crime, as the former can immediately provide cash (Wright and Decker 1997). Moreover, risky retailers tend to attract illegal activities, which often leads to violence and aggravated assault (Ousey and Lee 2002; Martinez, Rosenfeld, and Mares 2008). Given that dollar stores—although they do not sell inherently noxious or dangerous items like many other risky retailers—have also been reported to attract illegal drug trade (Klingseis 2014; MacGillis 2020) and their customers use cash as a frequent payment method, the effect of dollar stores on violent crime may be similar to that of risky retailers.

Despite numerous studies empirically investigating the relationship between various types of land use and crime, none have specifically investigated the impact of dollar stores on violent crime. However, with the massive growth of the dollar store industry in recent decades, which shows an unequal spatial distribution that may reinforce existing class and racial inequality, it is important to thoroughly understand community costs, including the heightening of violent crime. Specifically, across the United States, the number of dollar stores has increased more than three times, outpacing other types of food retailers (Chenarides et al. 2021). Moreover, it is widely known that dollar stores are disproportionally located in poor, ethnic minority neighborhoods (Shannon 2021). As this study shows, Chicago is no exception: the recent expansion of dollar stores mostly occurred in less affluent neighborhoods with a higher ethnic minority resident share. Therefore, if dollar stores contribute to an increase in violent crime, this raises serious equity concerns.

Against this backdrop, I examine how dollar stores affect localized violent crime patterns in Chicago using the historical Supplemental Nutrition Assistance Program (SNAP) retailer locator and crime data. Specifically, exploiting spatial and temporal variations in the openings or closings of new dollar stores, I analyze how the number of violent crimes in the store's immediate vicinity relative to control areas has changed after their opening or closing. I further explore whether these effects differ by crime location type and whether there is evidence of spatial displacement of crime. Finally, I investigate how the effects of dollar store openings evolve over time and how they vary according to the surrounding context of their locations to draw more detailed policy suggestions.

Literature Review

Various theories have been developed by criminologists and urban studies scholars to understand the geographic patterns of crime, some of which may elucidate the potential relationship between dollar stores and violent crime. These theories often complement one another, although their predictions regarding the impact of dollar stores on violent crime are somewhat mixed.

The broken windows theory posits that disorder in neighborhoods signals potential offenders that crime intervention social infrastructure is lacking, thus attracting more, especially serious, criminal activities (Wilson and Kelling 1982). Dollar stores are often seen as a symbol of disorder in surrounding areas not only due to their unattractive appearance and poor maintenance (Kassab 2013; Higgs 2020) but also due to the message they carry regarding the neighborhood's racial and class identity.¹ That is, the presence of dollar stores signifies poor ethnic neighborhoods, which are associated with the perception of disorder (Sampson and Raudenbush 2004).

In broken windows theory, disorder spurs fear of crime and reduces informal social control, thus further increasing crime. Wilson and Kelling (1982), in their article introducing the broken windows theory, particularly pointed out that street robberies are more likely to occur in areas with higher levels of disorder, as robbers might think they are less likely to be caught in such areas due to the many panhandlers' intimidating passersby. Sampson and Raudenbush (1999) further claimed that disorder is more likely related to robbery and burglary than other types of crime, as potential offenders typically select targets based on visual cues; however, in their empirical tests, neighborhood disorder was found to be positively associated with robbery rates only.

Moreover, routine activity theory postulates that the likelihood of crime occurring depends on the convergence of three elements—motivated offenders, suitable targets, and the absence of capable guardians—in space and time (Cohen and Felson 1979). This theory has been widely adopted to study robberies and sexual crimes among others (Miró 2014). In this theoretical framework, the opening of dollar stores is expected to lead to the convergence of the three elements. Here, suitable targets are dollar store patrons and employees, while motivated offenders are likely the residents of or visitors to the surrounding area; notably, dollar stores tend to open in disadvantaged areas where potential offenders are more likely to live (Bernasco 2010). Dollar stores also tend to lack professional security guards or high-quality surveillance cameras that can effectively serve as capable guardians (MacGillis 2020; Meyersohn and Cohen 2020), as their business model aims to minimize operating costs (Grigsby-Calage, Mullally and Volpe 2021). Further, dollar

stores in urban neighborhoods tend to have more obsolete security technologies than those in rural areas (Vargas 2021). Therefore, the number of crimes targeting store customers or employees is expected to rise around newly opened dollar stores in urban neighborhoods, whereas their closure likely has the opposite effect. However, the possibility of dollar store customers becoming “eyes on the street” (Jacobs 1961), acting as guardians providing natural surveillance, thus deterring the occurrence of crime, cannot be ignored.

These theoretical arguments, especially those related to routine activity theory, can be further developed by the literature on risky retailers and crime. Risky retailers, such as payday lenders (Kubrin et al. 2011) and alcohol outlets (Furr-Holden et al. 2016), are often characterized as crime-prone businesses with a spatially concentrated pattern. One common feature between risky retailers and dollar stores is their association with cash economies, as a relatively large portion of their transactions involves cash (Kubrin and Hipp 2016). This line of literature suggests that not only dollar store customers but also store employees who often manage cash registers and take cash deposits to the bank are more likely to be attractive targets for motivated offenders. Therefore, even if the store location was previously occupied by another retailer, dollar store openings are expected to heighten robbery in nearby areas. Given that a robbery provides cash instead of goods and can be done more quickly than breaking into buildings, offenders who target businesses associated with cash economies and their customers—those who probably have a pressing need for cash (e.g., for their drug purchase)—would prefer robbery to burglary or other types of property crime (Wright and Decker 1997).

The literature on risky retailers and crime further suggests another potential mechanism by which dollar stores increase violent crime rates—the rise in illegal activities, which often leads to violence. Anecdotes and media reports (Klingseis 2014; MacGillis 2020) suggest that dollar stores, like other risky retailers, attract drug trafficking due, in part, to the lack of security, which likely spurs various types of violent crime. Specifically, given the illicitness of their activities, drug market participants cannot rely on the police to protect them during their business transactions. Consequently, they often resort to firearms to resolve disputes that occur during the drug trade (e.g., buyers’ and sellers’ disputes over price and purity, turf battles between drug dealers), which results in aggravated assault and even homicide (Ousey and Lee 2002; Martinez, Rosenfeld, and Mares 2008). Additionally, drug dealers with both cash and drugs are good targets for robbers, especially those who rob to finance their drug purchases (Wright and Decker 1997). Ample empirical evidence confirms the theoretical prediction that risky businesses escalate violent crime in surrounding neighborhoods (Furr-Holden et al. 2016; Kubrin and Hipp 2016).

While no empirical research investigated the impact of dollar stores on crime, several studies have examined the effects of convenience stores—which are arguably similar to dollar stores in some respects, such as small square footage and limited security resources—on crime, generally reporting a positive association (Schweitzer, Kim, and Mackin 1999; Askey et al. 2018). For example, Furr-Holden et al. (2016) showed that the number of violent crimes decreases with the increasing distance from convenience stores in Baltimore.

However, dollar stores have distinct characteristics from convenience stores. Specifically, dollar stores rarely stay open late in the night or early in the morning, whereas many convenience stores are open 24 h a day, which puts the latter at a higher risk of crime. Moreover, dollar stores' primary customer base tends to be older and of lower income than those of convenience stores (Howell 2000; Shrestha 2016). The most common type of convenience store customer is a young male—a type more likely to be a violent crime victim (Perkins 1997). Furthermore, although selling alcohol is common at convenience stores, major dollar store chains did not start selling alcohol in many states until the late 2010s. Therefore, these differences imply that convenience stores may have a different, likely greater, impact on surrounding violent crime levels than dollar stores, highlighting the need for a dollar store-specific study.

Studies suggest that the effects of a certain type of land use or institution on nearby crime might be influenced by the surrounding (especially socioeconomic) context (Teh 2008; Stucky, Ottensmann, and Payton 2012). However, it is unclear how the socioeconomic status of surrounding neighborhoods moderates the impact of dollar store openings or closures on nearby violent crimes. On the one hand, dollar store openings may have less of an effect on violent crime in areas with lower socioeconomic status, as disadvantaged areas already have many other crime attractors.

On the other hand, dollar stores in wealthier areas may have less impact on violent crime, partly because they tend to have better management and appearance than those in traditionally disadvantaged areas. Reich (2016) coined the term “consumer redlining” to illustrate the phenomenon of Walmart stores located in lower-income ethnic neighborhoods having poorer conditions than those in wealthier neighborhoods; this phenomenon was also found in dollar stores (Vargas 2021). Additionally, in line with several theories such as social disorganization theory, higher levels of informal social control in wealthier neighborhoods may further mitigate the impact of dollar stores on violent crime. Given this theoretical ambiguity, exploring whether dollar stores similarly affect violent crime across different area characteristics can be an empirical question. Therefore, this study investigates how the relationship between dollar stores and violent crime varies depending on the surrounding context.

Furthermore, previous studies have often explored whether risky retailers shift criminal activity from one area to its vicinity, rather than increasing it. If this is the case, then the overall increase in crime at the city level would likely be modest. To date, research on risky retailers and crime has yielded mixed findings regarding such displacement effects (Teh 2008; Burkhardt and Goemans 2019). This study, therefore, aims to delve further into this issue in the context of dollar stores.

Data and Methodology

Data Sources

Dollar store data was primarily sourced from the U.S. Department of Agriculture's SNAP retailer database, which lists all SNAP-authorized retailers, including major dollar store chains (e.g., Dollar General, Family Dollar). As these chains have accepted SNAP benefit payments since the mid-2000s (Grigsby-Calage, Mullally, and Volpe 2021), dollar stores opened after this period are included in this database. Further, Shannon (2021) reported that since 2010, the number of dollar stores in the Reference USA commercial database—a frequently used employment database among urban researchers—has been similar to that in the SNAP data. Therefore, the study period was set from 2010 to 2019. Dollar stores were identified using store names containing “dollar” or “cent” (e.g., 99 Cents Only stores). A subsequent manual review confirmed whether most sale items were priced under \$10; this reflects the definition of dollar stores, as used in several local government ordinances.

Although the SNAP retailer database only has information regarding the starting and ending dates of the SNAP authorization, there might be stores that opened before 2010 but started accepting SNAP benefit payments during the study period. Therefore, I further employed the Chicago food inspection data² to check data quality. As dollar stores are required to have inspections of their establishments performed by the city's Department of Public Health to obtain a business license, I compared the SNAP authorization date to the “license” inspection date, finding them generally closer to one another. Similarly, the SNAP authorization end date was considered the date of store closure.

Concerns may also arise regarding the representativeness of the SNAP data, as stores in low-income areas are more likely to accept SNAP benefits. To explore potential selection biases, I compared SNAP data with data from Yelp (i.e., dollar stores listed on Yelp), a prominent online platform where consumers review and rate local businesses. Research has demonstrated that businesses in low-income or predominantly black neighborhoods tend

to receive fewer Yelp reviews or remain unlisted (Perry, Rothwell, and Harshbarger 2020; Shin, 2023; Telep and Hibdon 2023). Therefore, this bias suggests that dollar stores identified through Yelp may highlight selection issues that contrast with those in the SNAP data.

Figure A1 shows that the spatial distribution of dollar stores from both datasets (base year: 2022) is largely consistent, although SNAP contains more stores than Yelp. Comparing the median household income of census block groups with at least one dollar store between the two data reveals no significant difference: \$57,000 (SNAP) versus \$59,500 (Yelp), suggesting limited selection biases in the SNAP data.

Meanwhile, when a new dollar store opened at the same street address as another that had closed within the past year,³ it was considered the same entity, as there is no evidence to suggest that such a change affects criminal activity. Considering the proximity in timing, the launch of a new dollar store is not likely perceived as a genuine new opening but rather as a continuation of the former operation. With this consideration, I constructed the quarterly panel data for dollar stores that opened between 2010 and 2019; those that opened before 2010 were excluded from the main analyses.

Table A1 shows the basic descriptive statistics related to the number of dollar stores over the study period. Overall, the early study period (right after the Great Recession) registered a sharp increase in the number of dollar store openings. Accordingly, the number of total dollar stores showed a great increase between 2010 and 2015, slowing down afterward.

Information on violent crime was extracted from crime incident report data, downloaded from the City of Chicago data portal. This dataset contains all crime incident reports generated by the Chicago Police Department between January 1, 2010 and December 31, 2019. It provides extensive information, including the time, date, and detailed crime type. As the data provide latitudes and longitudes for the most proximate addresses for crime incidents, *point* features indicating the locations of crime incidents were created using ArcGIS. In this research, homicide, robbery, rape, and aggravated assault were all included to measure the number of total violent crimes; however, rape and homicides were excluded from the analysis performed by violent crime type; the former tends to be heavily underreported, while the latter is very rare, resulting in minimal variations across observations. A total of nearly 300,000 crime incidents, on average, occurred each year in Chicago over the study period, of which 9.8% were violent crime incidents.

The crime data also provide the category of location where each crime incident occurred, such as street and residence. Using this category, I distinguished between crime incidents occurring inside dollar stores⁴ and those occurring near them, such as commercial parking lots, and additionally performed a separate regression analysis by crime location type. Notably, the

location category was used only for this additional analysis by location type, while other analyses include all violent crime incidents, regardless of location category.

American Community Survey (ACS) data were also used to construct area socioeconomic and demographic variables in the regression analyses, which were performed using ArcGIS.⁵ Since this data is only available in five-year pooled estimates at the census block group level, I assigned the same values from 2010 to 2014 ACS data to individual quarters from 2010 to 2014, and the values from 2015 to 2019 ACS data to quarters from 2015 to 2019. Similarly, retail employment density was constructed using the annual Longitudinal Employer-Household Dynamics data.

Descriptive Statistics

This section discusses the results of descriptive statistics related to the location patterns of dollar stores. Preexisting dollar stores (opened before 2010) that were operational for at least part of the study period were also included, although they were distinguished from those that opened during the study period (stores included in the main analysis). This was done to check for any noticeable changes in the location pattern of dollar stores over time.

Table 1 reports the results of t-tests checking the statistical significance of mean differences in the major socioeconomic/demographic characteristics and violent crime densities of census block groups by dollar store presence. Significant differences were found in some characteristics between block groups with and without dollar stores. Specifically, block groups with either preexisting or recently opened dollar stores (i.e., stores opened between 2010 and 2019) tended to have lower socioeconomic status, as shown in lower median household income. However, the share of households below the poverty level was significantly higher only for block groups with preexisting stores than those without dollar stores. Furthermore, block groups with recently opened dollar stores showed different racial/ethnic compositions than those with preexisting dollar stores: while the latter tend to be located in block groups with a significantly higher proportion of Black residents, recently opened dollar stores seem to be located in block groups with a significantly higher Hispanic resident share. This may reflect the changing type of target customers as the dollar store industry sector expands its market. However, there is no significant difference in population density, regardless of the presence of dollar stores.

Table 1 further shows that violent crime density in 2009 was, on average, significantly higher for block groups with either preexisting or recently opened dollar stores than for those without. This is consistent with Figure 1 (left panel), which visually shows dollar store locations and violent crime

Table 1. Average Sociodemographic and Violent Crime Characteristics by Dollar Store Presence.

	CBGs without dollar stores (1)	CBGs with preexisting dollar stores (2)	CBGs with recently (2010–2019) opened dollar stores (3)	Difference ^a	
				(2)–(1)	(3)–(1)
Socioeconomic and demographic characteristics					
Population density (/acre)	27.98	26.83	30.51	–1.15	2.53
Median household income (\$)	50049.51	40615.36	41445.64	–9434***	–8603***
% Non-Hispanic blacks	40.47	50.23	45.31	9.76**	4.84
% Hispanic	23.03	26.36	32.87	3.33	9.84***
% Households below poverty level	19.90	22.99	21.77	–3.08**	–1.87
Crime					
Violent crime density (annual count/acre)	0.36	0.58	0.55	0.22***	0.19***
N	2,126	123	143		

Note. CBG = census block group. N indicates the number of CBGs. Socioeconomic and demographic averages are from 2005 to 2009 ACS data, while crime stats are from 2009.

^aThe column displays t-test results for differences in CBG characteristics and violent crime densities by dollar store presence: * $p < .10$, ** $p < .05$, *** $p < .01$.

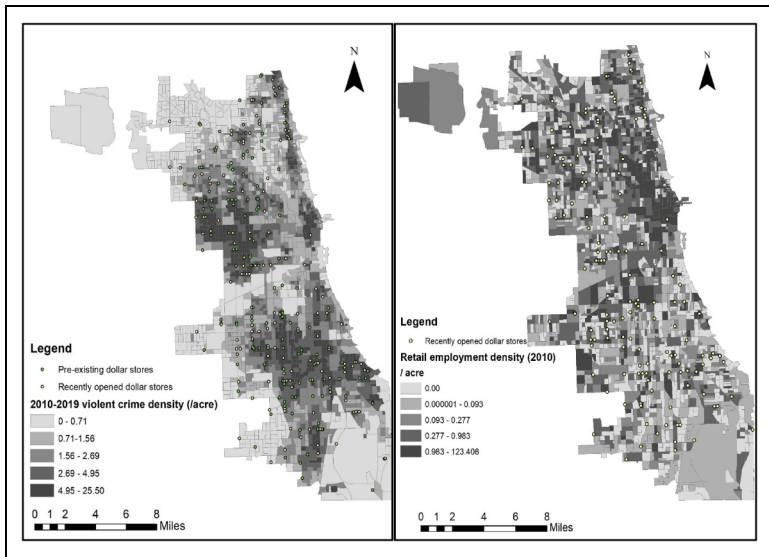


Figure 1. Dollar store locations with violent crime density (left) and retail employment density (right) at the census block group level in Chicago.

patterns during the study period. The study area has two large spatial clusters of violent crimes—the south and northwest sides of Chicago—where dollar stores are also concentrated. Therefore, these results indicate that the locations of dollar stores are not randomly distributed and are spatially overlapped with traditionally disadvantaged areas with a high initial level of violent crime.

Figure 1 (right panel) also compares the spatial distributions of dollar stores and retail employment density. Unlike retail in general, which tends to be concentrated near the city center, dollar stores are spatially more dispersed and tend to be located further from the center. A correlation between dollar store density and retail employment density at the block group level is also found to be low (-0.007) and statistically insignificant.

Methodology

Baseline Analysis. In this study, I employed a difference-in-differences (DID) approach to examine how the levels of violent crime change near dollar stores after store opening or closure. As Table 1 reveals significant differences in violent crime levels and other characteristics between census block groups with and without recently opened dollar stores prior to store opening, it is not ideal to use them as treatment and control groups, respectively.

Moreover, using a relatively large geographic unit (e.g., census tract or block group) for analysis can raise concerns of reverse causality when there is a strong spatial correlation between variables of interest at the macro geographic level (Cui and Walsh 2015). Given the overall spatial overlap between dollar stores and violent crime density found in Figure 1 (left panel), I utilized localized treatment and control areas.

Specifically, the treatment area was defined as a 250-foot buffer of each dollar store that opened between 2010 and 2019 (hereafter, “treatment ring”); the 250-foot buffer was chosen because it is approximately the size of a typical street block in Chicago (Heramb 2007). The “control ring” was defined as a doughnut circumscribing each treatment ring. A concentric ring has been employed to create controls that are comparable to treatment rings in several crime studies (Cui and Walsh 2015; Spader, Schuetz, and Cortes 2016). The advantage of this method is that the control and treatment rings are highly likely to resemble, in terms of both prestore opening crime levels and other contextual characteristics, due to their spatial proximity. Even within a census tract or block group⁶ that has been frequently used as a unit of analysis in crime literature, there are considerable variations in crime levels.

As shown in Figure 2, the control ring expands between 250 feet and 354 feet from each dollar store. A 354-foot cutoff was selected to define a control ring to make it equal to the treatment ring in terms of size; this makes it straightforward to compare the number of violent crimes between these two

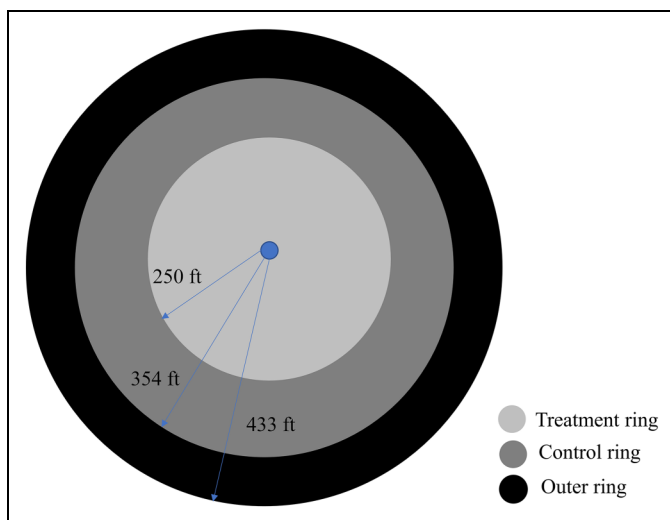


Figure 2. Definitions of treatment, control, and outer rings.

rings. The spatial unit of analysis in the present study is a ring surrounding each recently opened dollar store (opened between 2010 and 2019); therefore, the total number of observations in the baseline analysis reported in Table 3 is 12,080 (= 151 dollar stores × 2 rings [i.e., control, treatment] × 40 quarter-years).

In addition to the parallel trend assumption tests for crime outcome variables, which will be discussed later, I compared the observable characteristics of these two ring types to ensure the validity of the control ring. These characteristics are factors previously identified in research as influential on violent crime rates. Table 2 illustrates that, apart from a slightly higher median household income in the treatment rings, both treatment and control rings exhibit largely similar observable characteristics prior to the opening of dollar stores. Notably, although dollar stores are often located along main roads with other stores, no significant difference exists in commercial activity, measured by retail employment density, between the treatment and control rings.

The general form of the baseline DID analysis is shown in Equation (1):

$$\begin{aligned} \text{ViolentCrime}_{it} = & \alpha + \beta_1 \text{Open}_{it} + \beta_2 \text{Close}_{it} + \beta_3 T_r + \beta_4 T_r * \text{Open}_{it} \\ & + \beta_5 T_r * \text{Close}_{it} + \beta_6 Q_t + \beta_7 \text{Dollarstore}_i + \beta_8 \text{Control}_{it} + \varepsilon_{it} \end{aligned}$$

(1)

Table 2. Average Area Characteristics Between the Treatment and Control Rings Prior to Store Opening.

	Treatment ring	Control ring	
	(< 250 feet)	(250–354 feet)	Difference
	(1)	(2)	(1)–(2)
Area characteristics			
Median household income (\$)	40077.54	38766.61	1310.93***
% Non-Hispanic blacks	44.17	44.14	0.03
% Hispanics	34.44	34.37	0.07
% Households below the poverty level	24.84	25.08	–0.24
% Males aged 15 to 29	10.76	10.75	0.01
% Vacant housing units	15.64	15.59	0.05
Retail employment density (/acre)	0.86	0.86	–0.01
N	2627	2627	0

Notes. N indicates the sum of the number of pre-store opening quarters of dollar stores included in the analysis.

*p < .10, **p < .05, ***p < .01.

where $ViolentCrime_{itr}$ represents the number of violent crimes or their subtypes that occurred in quarter t for ring r of dollar store i ; $Open_{it}$ equals 1 if the dollar store i opened or was operational in quarter t and equals 0 otherwise; $Close_{it}$ equals 1 if dollar store i shut permanently before quarter t and equals 0 otherwise; T_r is a treatment ring indicator variable; $T_r * Open_{it}$ generates a coefficient estimate measuring the impact of opening a dollar store on violent crime levels in the treatment ring, relative to the control ring; $T_r * Close_{it}$ produces a coefficient estimate capturing the impact of a store closure; Q_t represents the quarter-year fixed effects; and $Dollarstore_i$ represents the dummy variables of 150 dollar stores, which adjust for time-invariant characteristics associated with each dollar store. By including these dollar store fixed effects, the regression model compares the crime levels in the treatment ring with those in the control ring for the same dollar store. Finally, $Control_{itr}$ includes median household income, percentages of non-Hispanic blacks, Hispanics, households below the poverty level, males aged 15 to 29, and vacant housing units, as well as retail employment density.

As the crime outcome variable is a count variable, a negative binomial regression model was used to estimate Equation (1), which is more appropriate for handling overdispersion than a Poisson regression model. Equation (1) was chosen as the baseline framework because it enables an analysis of the potential asymmetry in the effects of store openings versus closings. This was favored over the alternative specification, which does not distinguish between the preopening and postclosing periods and is concerned solely with the store's operational status.

Robustness Checks. I conducted several analyses to validate the DID design and ensure the robustness of the baseline results. First, although a 250-foot buffer is approximately equal to a typical block size in the study area, some might argue that the choice of size is arbitrary. Given that the exact point (in space) at which the impact of dollar stores on nearby crimes becomes insignificant is unknown, the effects of dollar store openings or closures on violent crimes may exceed 250 feet. To test whether the control group might be influenced by a dollar store opening or closing, I conducted additional analyses based on Spader et al.'s (2016) methodology. Specifically, I created an "outer ring" that surrounds each control ring and falls between 354 and 433 feet (see Figure 2); the size of the outer ring is also the same as that of the other two rings to simplify calculations. In this approach, both the treatment and control rings are considered potential treatment groups while the outer rings serve as control groups. Therefore, if the effect of a dollar store opening or closing goes beyond 250 feet, there would be significant differences in post-store opening or closing crime levels between the control and the outer rings.

Second, a key identifying assumption of the DID analysis is the pretreatment parallel trend. To verify this assumption, I employed the event study approach, following previous literature (He, Pan, and Tanaka 2020; Chang and Diao 2022). This approach is an extension of a DID analysis in which the interaction term between the treatment ring status and poststore opening period ($T_r * Open_{it}$ in Equation 1) is replaced by a vector of interaction terms between a treatment ring status variable and individual quarters immediately before and after dollar store openings. I considered eight lags ($t-8$ or earlier to $t-1$) and 13 leads ($t+1$ to $t+13$ or later) relative to the store's opening quarter (t), with one quarter before the store opening serving as the reference. If the treatment and control rings have parallel trends in crime levels in the quarters prior to dollar store openings, the results of the event study analysis should indicate statistically insignificant coefficients for the interaction terms between the treatment ring status and each lag variable. Recognizing that certain stores both opened and closed during the study period, the analysis period was narrowed down to consider quarters before store closures for these establishments.

Third, another methodological concern is the potential violation of the assumption in DID analysis that each observation should be independent. This is because some dollar stores are close to one another, which leads to spatial overlaps between some treatment and control rings. Not only can such overlaps affect the magnitude of standard errors, but they likely bias estimation results downward. To address this concern, I conducted additional analyses using only dollar stores without nearby (< 604 feet) dollar stores; this distance of 604 feet represents the sum of 250 feet and 354 feet.

Fourth, recent econometric studies have highlighted potential biases in the conventional DID approach with entity and time-fixed effects. Significant biases arise, especially when treatment times vary and when uneven treatment effects occur over time. This bias is rooted in the setup of the conventional model, where groups that have previously received treatment are used as controls in scenarios with multiple treatment periods. Given the phased openings of dollar stores, this issue is relevant. To address this concern, I adopted Callaway and Sant'Anna's (2021) staggered DID model, henceforth CS DID.

One fundamental assumption of the CS DID is the irreversibility of treatment (i.e., the treatment is permanent and cannot be undone). Given that some dollar stores both opened and closed during the study period, the CS DID estimation solely concentrates on the effects of store openings, after restricting the samples to areas (rings) associated with dollar stores that opened and subsequently remained open.⁷ Specifically, the CS DID model estimates the average treatment effect on the treated (ATT) using only clean comparisons, which is particularly suited for scenarios involving multiple treatment groups spanning different periods. In this study, treatment rings surrounding dollar

stores that opened within the same quarter belong to the same “group.” Clean comparisons are made by considering only units that have never been treated as the control group. The CS DID model permits the aggregation of the disaggregated parameters into several different measures of causal effects, including group-time aggregates, time average, and group average. The CS event study plot was also provided to visualize the treatment effect dynamics.

Additional Analysis. Several additional analyses were performed to draw more specific and valuable policy implications, including a separate regression analysis by type of crime location (e.g., parking lots, streets/sidewalks). I aimed to determine whether the effects of dollar store openings and closures are confined to the store building; if not, interventions also need to focus on these affected areas.

Another crucial issue related to the research question is whether changes in crime rates, following the opening or closing of a dollar store, reflect a mere spatial displacement of crime. For example, if violent crimes simply move from nearby areas to within 250 feet of a dollar store after its opening, the overall impact on the citywide crime rate could be modest. Following prior crime literature (Telep and Hibdon 2018; Hatten and Piza 2022), I refined the method used to examine the spatial extent of the impact of dollar store openings by devising comparison rings (250-foot buffers) outside the areas potentially impacted by dollar stores (see Figure A2). Each dollar store was paired with one comparison ring, which was selected through the following criteria:

1. The centers of comparison rings lie beyond 683 feet⁸ from any dollar store, yet within a radius of four street blocks from their respective corresponding dollar store.
2. Many dollar stores are located on major, high-traffic roads, which can influence crime levels; accordingly, the centers of the comparison rings are situated on roads of a similar hierarchy, or possibly on the same road.
3. The land use patterns in the comparison rings match those near their corresponding dollar store.

The centers of comparison rings were manually selected using OpenStreetMap and ArcGIS. Subsequently, I extended the baseline DID framework variables by adding those related to control and outer rings, including interaction terms between these rings and the store’s opening or closing status. This approach enabled an analysis of how dollar store openings or closures affect violent crime within treatment (within 250 feet), control (250–354 feet), and outer (354–433 feet) rings, as compared to new comparison rings. Thus, in this setting, the control and outer rings were viewed as potential buffer zones, where crime could shift toward the treatment rings from these areas, or vice versa.

Finally, I further investigated the heterogeneity of effects that dollar store openings have on nearby violent crimes across surrounding area characteristics, using a machine learning approach, specifically the causal random forest (CRF) developed by Wager and Athey (2018). The CRF, increasingly used in social science research to explore treatment effect heterogeneity (see Farbmacher, Kögel, and Spindler 2021; Weber and Cappellari 2022), offers several advantages. Notably, it excels in capturing complex, nonlinear relationships between variables and automatically identifying subgroups that are most affected by the treatment.

This approach is used to estimate the conditional average treatment effect (CATE), which represents the expected difference in outcomes between treated and untreated groups, conditional on specific covariates. In the CRF method, each tree in the forest is constructed using a bootstrap sample, and a subset of covariates is selected for analysis. Unlike traditional RFs, trees in CRF are grown by identifying splits that maximize the heterogeneity in treatment effects between the child nodes.

In this study, the analysis consists of two distinct stages. Initially, I determined the covariates that have the greatest impact on the heterogeneity of treatment effects. This was achieved by utilizing a variable importance measure, which quantifies how frequently the forest splits on specific covariates across all the trees it contains. Essentially, this measure identifies the variables that are most likely to contribute to variations in the CATEs. Next, to investigate how the effect on crime differs across these variables, I created heatmaps presenting how the CATEs change across the two most important variables while holding other covariates at their average values. This analysis examined the heterogeneous impact of store openings only, relying on the restricted sample set used for the CS DID. This focus is owing, in part, to the limited number of stores that closed, as it is insufficient for conducting a robust machine learning analysis.

The CRF analysis was performed using the *grf* package in R. For each specific crime outcome variable, I performed a separate CRF. I grew each forest using 5,000 trees, ensuring that each leaf node contained a minimum of 15 observations. To maintain consistency with the default settings of the *causal_*-*forest* function in R, I constructed each decision tree using a 50% random subsample of our analysis dataset, adhering to the honest approach.

Results

Baseline Results

Table 3 provides the baseline estimates of the impact of dollar store openings or closures on nearby violent crimes. Each outcome variable has two

Table 3. Baseline Results: Negative Binomial DID Estimation Results.

Outcome	Violent crime		Robbery		Aggravated assault	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment Ring × Store Opening	0.193* (0.098)	0.195** (0.098)	0.321*** (0.117)	0.322*** (0.116)	0.127 (0.115)	0.132 (0.115)
Treatment Ring × Store Closure	0.111 (0.200)	0.131 (0.177)	0.014 (0.264)	0.044 (0.225)	0.154 (0.200)	0.179 (0.179)
Store opening	-0.140* (0.077)	-0.137* (0.076)	-0.171* (0.095)	-0.162* (0.094)	-0.147* (0.088)	-0.150* (0.088)
Store closure	-0.167 (0.121)	-0.181 (0.117)	-0.074 (0.155)	-0.066 (0.148)	-0.212* (0.125)	-0.250** (0.121)
Treatment ring	0.106 (0.089)	0.103 (0.088)	0.171* (0.092)	0.165 (0.100)	0.053 (0.107)	0.053 (0.107)
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Store fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Control variables	No	Yes	No	Yes	No	Yes
Mean crime (per quarter)	0.80		0.39		0.38	
Observations	12,080		12,080		12,080	

Notes: Observations = 12,080 (= 151 Stores × 2 Rings × 40 Quarters). Numbers in parentheses are clustered standard errors. Control variables include median household income, percent non-Hispanic blacks, percent Hispanics, percent households below poverty level, percent males aged 15 to 29, percent vacant housing units, and retail employment density. DID = difference-in-differences.

* $p < .10$, ** $p < .05$, *** $p < .01$.

columns: one with control variables and the other excluding them. The results are similar, regardless of the inclusion of control variables. Column 2 indicates that areas near dollar stores experience an overall increase in violent crime levels after store openings. Specifically, the treatment ring sees an average increase of approximately 0.17 more violent crimes per quarter⁹ relative to the control ring, translating into an increase of about 21.5% relative to the mean. Moreover, the interaction coefficient between the treatment ring and the store closure status indicates that, after the closure of dollar stores, violent crime levels around stores relative to the control rings are not statistically different from those of the prestore opening period. As some properties became empty while others transitioned to different businesses after dollar stores' closure, this result can be understood as a combined average effect of both scenarios.

The results of the regression analysis, broken down by crime subtype, are shown in columns 3–6. Interaction coefficients between the treatment ring and the store opening status indicate that the overall rise in the number of violent crimes near stores after their opening is primarily driven by an increase in robberies. This is because the opening of new dollar stores does not have a significant impact on aggravated assault rates in the treatment ring relative to the control ring, while it is positively associated with robbery rates. Specifically, the treatment ring experiences approximately 0.15 more robberies per quarter compared to the control ring, which corresponds to an increase of roughly 38.0% relative to the mean. These results reveal the heterogeneous impact of dollar store openings on different violent crime types.

While positive, none of the coefficients on the interaction between the treatment ring and the store closure status are statistically significant. This suggests that, regardless of crime type, the levels of violent crime in the treatment ring relative to the control ring revert to levels comparable to those before the store opening once the stores permanently close. That is, although some residual effects of dollar store openings on nearby violent crime rates persist, the magnitude of these effects is statistically negligible.

Robustness Check Results

Spatial Extent of the Dollar Store's Impact. Table 4 presents the results of the additional analyses on whether the impact of dollar store openings or closures on violent crimes extends beyond 250 feet. As explained earlier, the study area was expanded from 354 to 433 feet from dollar stores, and control ring-related variables were added to the baseline models. That is, I analyzed the violent crime level changes in the treatment and control rings against the outer rings (354–433 feet from dollar stores) post dollar store openings and closures.

Table 4. Negative Binomial DID Estimation Results with Three Rings: The Addition of the Outer Ring.

Outcome	Violent crime	Robbery	Aggravated assault
	(1)	(2)	(3)
Treatment Ring $\times$ Store Opening	0.195* (0.114)	0.301** (0.124)	0.160 (0.136)
Control Ring $\times$ Store Opening	-0.001 (0.116)	-0.024 (0.133)	0.032 (0.135)
Treatment Ring $\times$ Store Closure	0.067 (0.118)	0.052 (0.141)	0.147 (0.139)
Control Ring $\times$ Store Closure	-0.042 (0.202)	0.039 (0.280)	-0.015 (0.184)
Quarter fixed effects	Yes	Yes	Yes
Store fixed effects	Yes	Yes	Yes
Control variables	Yes	Yes	Yes
Mean crime (per quarter)	0.73	0.34	0.36
Observations	18,120	18,120	18,120

Notes. $N = 18,120$ ($= 151$ Stores $\times 3$ Rings $\times 40$ Quarters). Numbers in parentheses are clustered standard errors. All models additionally include dummy variables for opening status, closure status, treatment ring, and control ring. Control variables are the same as those included in Table 3. DID = difference-in-differences.

* $p < .10$, ** $p < .05$, *** $p < .01$.

Overall, the statistical significance and signs of the key interaction coefficients related to the treatment ring (treatment $\times$ opening, treatment $\times$ closure) do not change compared to those reported in Table 3. Specifically, the numbers of violent crimes (total) and robberies relatively increase in the treatment rings compared to the outer rings after new dollar stores open. Moreover, after the stores' permanent closure, the levels of violent crime in treatment rings relative to the outer rings become insignificantly different from those of the prestore opening period, regardless of crime type.

Given that control rings serve as potential treatment areas in this additional analysis, the coefficients on the interaction between the control ring status and store opening/closure status are important to note. Results indicate no significant change in crime rates within control rings relative to outer rings following store opening or closure. These results imply that the effective treatment area is unlikely to exceed 250 feet from a dollar store. That is, there is a positive effect of dollar store openings on surrounding violent crime, but only in a very localized manner.

Parallel Trend Test. Figure 3 displays the results of the event study analysis, testing the parallel trend assumption of the DID analysis. Before dollar store openings, the trends of violent crime and its subtypes were consistent between treatment and control rings, as revealed by statistically insignificant coefficients for all lag variables across all panels of Figure 3. This confirms the suitability of the DID methodology for this study. Figure 3 further shows a rise in the levels of total violent crime and robbery in the treatment rings relative to the control rings, becoming more evident in a later period following the openings of dollar stores. However, this pattern is less pronounced for aggravated assaults.

Spatial Proximity of Dollar Stores. Table A2 presents the results of DID analyses conducted after excluding dollar stores neighboring other dollar stores (located within 604 feet of other dollar stores). These extra analyses help determine whether a close distance between some dollar stores and the resultant spatial overlap between some control and treatment rings have a critical influence on baseline findings. Table A2 indicates that the results of the baseline analysis are retained. Specifically, the positive effects of dollar store openings on violent crime overall and robbery incidents remain statistically significant, despite the reduced sample size; however, as expected, the magnitude of these effects is larger in Table A2 than in Table 3. Therefore, these results suggest that the true effects of dollar store openings on violent crime may be slightly larger than the baseline results.

CS DID Model and Long-Term Effects. Table 5 displays the CS DID results, which generally align with the baseline results in Table 3. The group-time ATTs highlight significant rises in total violent crimes and robberies in areas immediately surrounding a newly opened dollar store. By contrast, the number of aggravated assaults remains unchanged postopening. Regarding the disaggregated group-specific treatment effects, a clear variation exists in the impact of store openings across different groups; while many coefficients are positive, some exhibit statistically significant negative values. However, positive group-specific coefficients outnumber negative ones for total violent crimes and robberies, which likely influences group average estimates.

In addition to the preaverage and postaverage ATT reported in Table 5, Figure 4 graphically presents the dynamic treatment events based on the event study plots. Both results show no significant pretreatment trends for any outcome variables. By contrast, the posttreatment trends align with the baseline results, showing the positive effects of store openings on violent crimes and robberies during this period. Furthermore, while some fluctuations

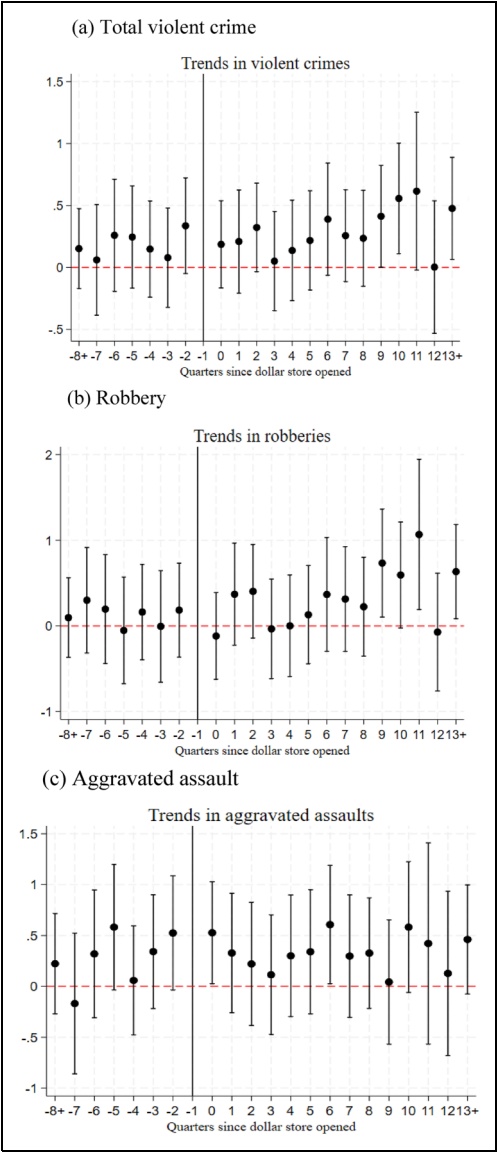


Figure 3. Event study results: pre- and poststore opening trends in violent crime, robbery, and aggravated assault. (a) Total violent crime. (b) Robbery. (c) Aggravated assault.

Notes. These figures present the coefficient estimates along with their 90% confidence intervals. The notation “13+” (“8+”) refers to 13 (8) or more quarters after (before) the opening of the dollar store.

Table 5. CS DID Estimation Results.

	Violent crime		Robbery		Aggravated assault	
	Coef	S.E.	Coef	S.E.	Coef	S.E.
Group-time aggregate	0.161*	0.08	0.094*	1.82	0.062	0.08
Group average	0.094	0.07	0.093**	0.04	-0.006	0.05
Quarter 2	0.366***	0.09	0.357**	0.14	0.012	0.16
Quarter 3	0.059	0.53	0.214***	0.08	-0.138	0.49
Quarter 4	0.959***	0.13	0.638***	0.06	0.329***	0.12
Quarter 5	0.679**	0.35	0.261**	0.13	0.413*	0.22
Quarter 7	0.623***	0.13	0.125	0.18	0.478***	0.17
Quarter 8	-0.634***	0.10	-0.878***	0.07	0.231***	0.08
Quarter 9	0.126	0.37	0.311***	0.12	-0.207	0.41
Quarter 10	0.16	0.28	0.241***	0.04	-0.113	0.29
Quarter 11	0.106	0.09	0.059	0.06	0.011	0.06
Quarter 12	-0.407***	0.08	0.566***	0.05	-1.045***	0.06
Quarter 13	0.208	0.23	-0.147	0.32	0.334*	0.18
Quarter 14	0.851***	0.07	0.131***	0.04	0.765***	0.05
Quarter 15	0.144	0.10	0.111**	0.05	0.052	0.07
Quarter 16	0.689***	0.08	0.359***	0.06	0.354***	0.06
Quarter 17	-0.381***	0.09	-0.177	0.43	-0.189	0.43
Quarter 18	0.012	0.15	-0.195	0.17	0.221**	0.09
Quarter 19	0.023	0.10	-0.088	0.07	0.083	0.07
Quarter 20	0.178*	0.10	0.114**	0.06	0.091	0.06
Quarter 21	0.922***	0.08	0.422***	0.05	0.454***	0.05
Quarter 22	0.250	0.15	0.006	0.06	0.267***	0.10
Quarter 23	0.048	0.18	-0.275	0.28	0.318*	0.18
Quarter 24	-1.832***	0.06	-0.934***	0.04	-0.946***	0.05
Quarter 25	-0.208	0.31	0.108*	0.06	-0.282	0.31
Quarter 26	0.175**	0.08	0.088*	0.05	0.097**	0.05
Quarter 27	0.295**	0.13	0.368*	0.20	-0.054	0.28
Quarter 28	0.736***	0.10	0.180***	0.07	0.570***	0.06
Quarter 29	0.740**	0.31	0.448**	0.21	0.318**	0.13
Quarter 30	-0.395*	0.22	-0.042	0.06	-0.374*	0.22
Quarter 31	0.040	0.40	-0.026	0.24	0.081	0.17
Quarter 32	0.867***	0.29	0.331	0.28	0.514***	0.16
Quarter 33	-5.683***	0.08	-1.504***	0.06	-4.303***	0.04
Quarter 34	-0.269	0.38	-0.179	0.33	-0.065	0.38
Quarter 35	0.982	1.12	-0.119	0.53	1.079*	0.59

(continued)

Table 5. (continued)

	Violent crime		Robbery		Aggravated assault	
	Coef	S.E.	Coef	S.E.	Coef	S.E.
Quarter 36	0.425***	0.07	0.211***	0.04	0.200***	0.04
Quarter 37	-0.456	0.31	-0.001	0.09	-0.610***	0.28
Quarter 38	0.054	0.60	0.584**	0.24	-0.530	0.36
Quarter 39	-1.183	1.24	0.362*	0.19	-1.528	1.06
Quarter 40	1.567***	0.36	0.978	0.07	0.579	0.36
Time average	0.205**	0.09	0.144***	0.05	0.057	0.09
Event study aggregates						
Preaverage	-0.018	0.03	-0.0072	0.01	-0.009	0.03
Postaverage	0.277**	0.11	0.155***	0.06	0.118	0.09

Notes. $N = 7, 120$. DID = difference-in-differences.

* $p < .10$, ** $p < .05$, *** $p < .01$.

exist, the impact of dollar store openings on violent crimes and robberies increases over time, becoming more pronounced particularly several years following the openings. Although the overall trends for aggravated assaults seem similar to those for total violent crime and robbery levels, most coefficients are not statistically different from zero, even during the postopening period.

Additional Analysis Results

Analysis by Type of Crime Location. Table 6 presents the results of separate regression analyses performed by location type (i.e., streets/sidewalks, residences, parking lots, small retail/convenience stores, or other places) to identify where exactly violent crime is more likely to occur. Violent crime incidents disaggregated by location type were not further classified by crime type due to their relatively small frequency within many treatment or control rings.

Results reveal that the impact of new dollar store openings on violent crime is not limited to the store itself, but also exists on adjacent streets or sidewalks. Given the absence of a significant increase in violent crime in nearby parking lots, customers walking to and from dollar stores may be at a heightened risk of becoming violent crime targets. After a store's closure, the heightened crime rates observed within the store and on adjacent streets/sidewalks seem to revert to the levels seen prior to the store's opening. While the opening of dollar stores does not lead to an increase in

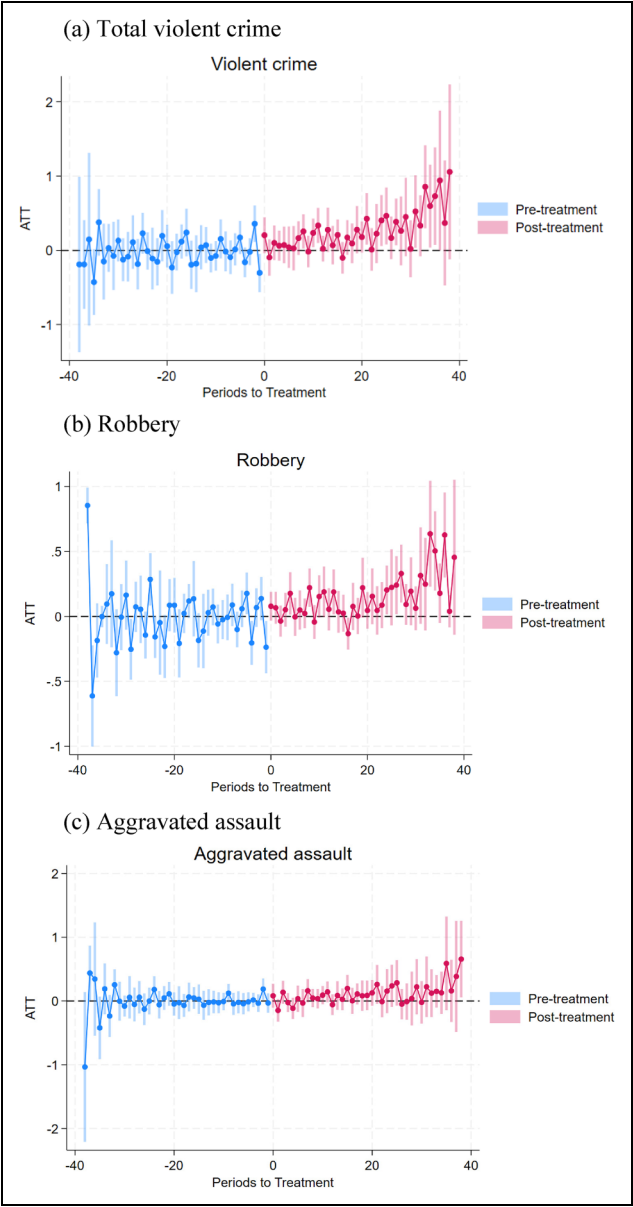


Figure 4. Dynamic treatment effects based on the CS DID. (a) Total violent crime. (b) Robbery. (c) Aggravated assault.
Note. These figures report the coefficient estimates and their 90% confidence intervals. DID = difference-in-differences.

Table 6. Negative Binomial DID Estimation Results by Type of Crime Location.

	Street/ sidewalk	Residence		Small retail/convenience store	Parking lot (nonresidential)	Other places
	(1)	(2)	(3)	(4)	(5)	
Outcome						
Treatment Ring × Store Opening	0.177* (0.104)	−0.329 (0.275)	0.637** (0.320)	0.045 (0.259)	0.072 (0.135)	
Treatment Ring × Store Closure	0.213 (0.141)	0.777*** (0.287)	−0.024 (0.317)	0.483 (0.450)	0.030 (0.274)	
Store opening	−0.178** (0.075)	−0.144 (0.204)	−0.022 (0.262)	−0.030 (0.252)	−0.107 (0.105)	
Store closure	−0.244*** (0.094)	−0.888*** (0.263)	0.171 (0.329)	−0.696 (0.462)	−0.077 (0.197)	
Treatment ring	0.003 (0.093)	−0.808*** (0.211)	0.846*** (0.249)	0.785*** (0.226)	0.060 (0.136)	
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	
Store fixed effects	Yes	Yes	Yes	Yes	Yes	
Control variables	Yes	Yes	Yes	Yes	Yes	
Mean crime (per quarter)	0.42	0.04	0.06	0.04	0.26	
Observations	12,080	12,080	12,080	12,080	12,080	

Notes. N = 12,080 (= 151 Stores $\times$ 2 Rings $\times$ 40 Quarters). Numbers in parentheses are clustered standard errors. Control variables are the same as those included in Table 3. DID = difference-in-differences.

* $p < .10$, ** $p < .05$, *** $p < .01$.

violent crimes in adjacent residential areas, the incidence of violent crimes in residences rises in comparison to control areas following the closure of these stores.

Spatial Displacement of Crime. Table 7 examines whether dollar store openings or closings influence violent crime through spatial crime displacement. The effects of dollar store openings on violent crime near the store (within 250 feet) align with the baseline findings in Table 3, regardless of crime type; compared to the comparison rings, treatment rings see significant increases in violent crimes and robberies after a dollar store opens. The coefficients for interactions between either the control or outer ring and store opening status are negative, but not statistically significant. This indicates that the rise in violent crimes and robberies within 250 feet of dollar stores postopening is not owing to the significant displacement of these crimes from nearby areas. Similarly, because the interactions of both the control and outer ring with the store closing status are not statistically significant, a return to preopening levels of violent crime in treatment rings after store closure does not suggest the displacement of crime from treatment rings to neighboring areas.

Heterogeneous Effects of Dollar Store Openings on Crime. Before examining the heterogeneity in the effects of dollar store openings on violent crime across different neighborhood characteristics, Table A3 displays the average treatment effects from the CRF analysis. These results align closely with those from the baseline DID analysis. Specifically, dollar store openings are found to have a significant and positive impact on overall violent crime and robbery, while their effect on aggravated assault is insignificant.

The variable importance measures reveal that the two important variables in determining treatment effect heterogeneity for violent crime outcomes are median household income and retail employment density, each with values exceeding 0.14, which is noticeably higher than those of other variables. Therefore, Figure 5 displays the variation in estimated effects of dollar store opening on each outcome variable in relation to median household income levels and retail employment density. Across both income and retail density spectrums, all outcome variables demonstrate positive CATE values. However, the relative magnitude of aggravated assaults is notably smaller, aligning with the baseline findings. Moreover, overall, Figure 5 indicates that the variation in CATEs is more strongly influenced by neighborhood income levels than by retail employment density.

Specifically, the influence of dollar store openings on robberies is notably more pronounced in lower-income neighborhoods compared to wealthier ones. Regarding total violent crimes and aggravated assaults, the effect is

Table 7. Negative Binomial DID Estimation Results for the Spatial Displacement of Crime.

Outcome	Violent crime	Robbery	Aggravated assault
	(1)	(2)	(3)
Treatment Ring × Store Opening	0.162* (0.098)	0.247** (0.111)	−0.007 (0.122)
Control Ring × Store Opening	−0.093 (0.086)	−0.078 (0.110)	−0.135 (0.103)
Outer Ring × Store Opening	−0.095 (0.090)	−0.052 (0.120)	−0.174 (0.107)
Treatment Ring × Store Closure	0.119 (0.144)	0.204 (0.128)	0.073 (0.187)
Control Ring × Store Closure	0.018 (0.204)	0.200 (0.257)	−0.082 (0.207)
Outer Ring × Store Closure	0.058 (0.166)	0.159 (0.192)	−0.060 (0.180)
Quarter fixed effects	Yes	Yes	Yes
Store fixed effects	Yes	Yes	Yes
Control variables	Yes	Yes	Yes
Mean crime (per quarter)	0.74	0.34	0.36
Observations	24,160	24,160	24,160

Notes. $N = 24,160$ ($= 151$ Stores $\times$ 4 Rings $\times$ 40 Quarters). Numbers in parentheses are clustered standard errors. All models additionally include dummy variables for opening status, closure status, treatment ring, control ring, and outer ring. Control variables are the same as those included in Table 3. DID = difference-in-differences.

* $p < .10$, ** $p < .05$, *** $p < .01$.

smaller in neighborhoods with a median household income below roughly \$23,000; however, in areas with income levels ranging from \$23,000 to \$50,000, the positive impact of dollar store openings on these crimes is more noticeable than in higher-income neighborhoods. Meanwhile, irrespective of income levels, areas with higher retail employment density experience a somewhat larger increase in overall violent crime and its subtypes following a dollar store’s opening. These results therefore provide a more nuanced picture of the impact of dollar store openings on violent crimes.

Discussion and Conclusions

Given the explosive growth of dollar stores over the past decade and their continued expansion during the COVID-19 pandemic, it is timely and

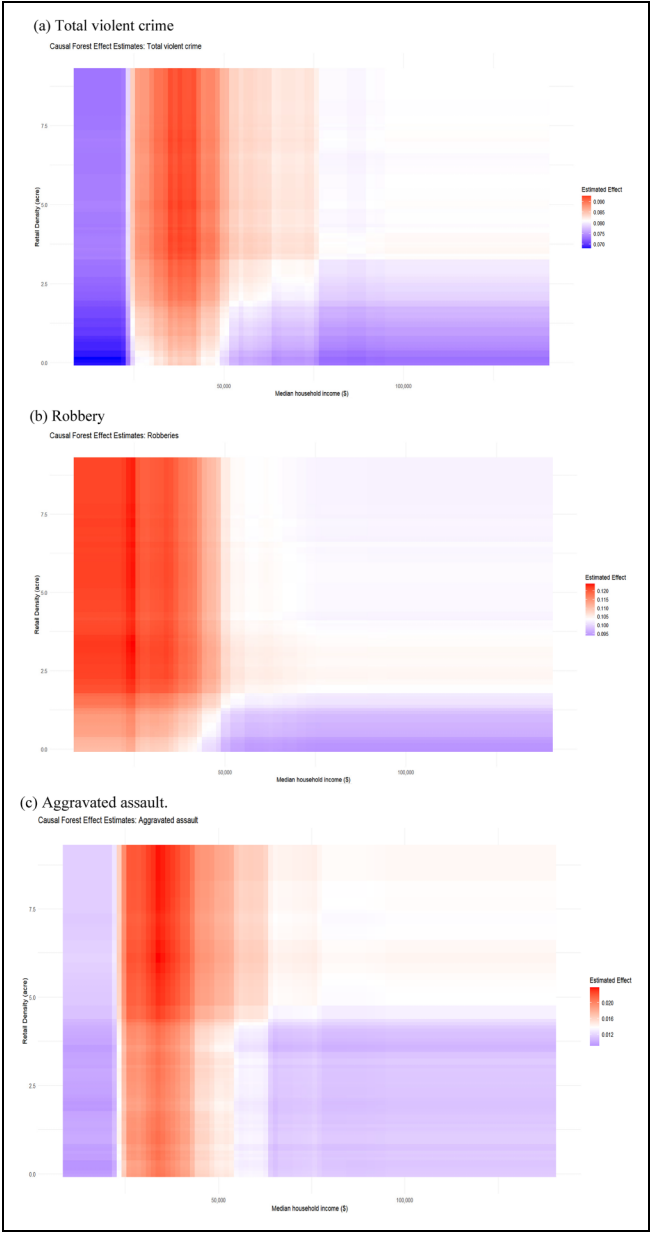


Figure 5. CRF estimates on median household income-retail employment density grid. (a) Total violent crime. (b) Robbery. (c) Aggravated assault. CRF = causal random forest.

crucial to understand how they affect surrounding communities. This study contributes to the literature by offering pioneering and rigorous evidence regarding the link between dollar stores and violent crime.

One important takeaway of this study is that dollar store openings are consistently associated with higher levels of violent crime overall, even after adjusting for various covariates. Specifically, there is, on average, a 21.5% increase in the number of total violent crimes after dollar store openings in their immediate vicinity relative to the control rings; although direct comparison is not possible due to different model specifications and study periods, this percentage is slightly larger than the effect of vacant foreclosed properties on violent crime in the study area (Cui and Walsh 2015). Importantly, this increase is unlikely the result of the spatial displacement of crime. Results further indicate that the levels of violent crime near dollar stores relative to the control rings return to those of the prestore opening period after stores shut permanently, which reaffirms the critical role played by dollar stores in local crime patterns.

Furthermore, the relationship between dollar store openings and violent crime is not uniform across crime types. As several theories suggest, including the routine activity and broken windows theories, the impact of dollar store openings is found to be more pronounced for robberies. This finding is similar to those of previous studies examining the link between risky retailers and violent crime (Kubrin and Hipp 2016; Feng et al., 2019). However, unlike other risky retailers, such as marijuana dispensaries (Contreras 2017) and fringe banks (Kubrin and Hipp 2016), dollar stores are not associated with aggravated assaults. Given this lack of association between dollar stores and violent crime other than robbery, a rise in illegal activities, such as drug trafficking, is less likely the mechanism through which dollar stores increase nearby violent crime.

These results also suggest that dollar store patrons are less likely to serve as informal surveillance ("eyes on the street"). Instead, they are in line with the idea that the lack of professional guardians and security cameras in or near dollar stores promotes potential offenders to rob store employees or customers, who are more likely to carry cash. Results also do not contradict the broken windows theory view in which dollar stores might be regarded as a neighborhood disorder, thus increasing violent crime levels. However, the present study does not elucidate which theoretical mechanism better explains the relationship between dollar stores and robbery; this can be further explored in future studies.

Another important takeaway is related to the spatial extent of the impact of dollar stores on violent crime. After their opening, the levels of violent crime rise not only inside stores but also on nearby streets or sidewalks. Although it is not possible to identify who the victims are, this indicates that customers

who visit dollar stores on foot are highly likely to be the targets of violent crimes, particularly robbery. However, such an increase in violent crime levels is less likely to go beyond 250 feet from dollar stores, suggesting that the effect of dollar stores is spatially microscopic.

Finally, the impact of dollar store openings on violent crime is dynamic, influenced both by time and the surrounding environment. The effect of store openings on violent crimes tends to intensify over time, particularly a few years after opening. Furthermore, neighborhoods with lower median household incomes, except for the lowest end of income areas, experience a greater increase in total violent crimes after a dollar store opens. Overall, these results mirror results from other risky retailers (Teh 2008; Burkhardt and Goemans 2019). Additionally, in areas with denser retail employment, a greater increase in total violent crime and its subtypes postopening is observed, regardless of income levels.

The results of this study are valuable for planners and policymakers in justifying the classification of dollar stores as crime attractors and implementing regulations limiting the number of dollar stores in close proximity. In recent years, a growing number of local jurisdictions have enacted legislation to restrict the licensing and operation of dollar stores (Bhattarai 2021), the most common justifications being the protection of retail choice diversity, community health, and public safety. However, most of these justifications have not been empirically substantiated.

Based on this study's findings, policymakers may consider requiring dollar stores to increase their security efforts in exchange for obtaining a business license, instead of controlling their density by mechanically setting distance limits between stores. This strategy has the dual benefit of directly targeting the issue of security while also potentially curbing the expansion of dollar stores by adding to their operational costs. For example, the adoption of new security technologies or the installation of high-quality security cameras in appropriate locations could be potential requirements for obtaining a business license. Policymakers may also consider pushing dollar stores to change their operating practices that seem to incur security issues, such as making employees take cash deposits to banks by themselves (Meyersohn and Cohen 2020).

The findings also underscore the need to channel more security measures and interventions toward vicinities of dollar stores located in lower-middle-income areas, especially in areas with a high concentration of retail outlets. Given that the influence of dollar store openings on violent crime grows over time, particularly several years postopening, it is essential to periodically review and not scale back the relevant security measures and policies. As dollar stores affect crime levels, both inside them and on nearby streets/side-walks, any policy designed to reduce crime in relation to dollar stores should not limit their geographic scope to store buildings.

Despite its crucial contributions to the literature, this study is not without limitations, which provide directions for future studies. First, the primary data used in this study lack information on the detailed characteristics of each dollar store, such as security systems and parking lot size, although previous studies have suggested that these play important roles in promoting or preventing crime (D'Alessio and StolzenBerg 1990; Casteel and Peek-Asa 2000). Therefore, further studies are needed to explore how the impact of dollar stores on violent crime differs by store characteristics. Second, the impact of dollar stores on local crime patterns might differ in other cities with different physical and social contexts than Chicago. Therefore, further examination of other cities or rural areas is needed to confirm the generalizability of this study's results.

Acknowledgments

The author would like to thank participants at the 2022 AAG annual meeting for their valuable comments.

Declaration of Conflicting Interests

The author declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was supported by Ministry of Education of Republic of Korea and National Research Foundation of Korea (BK21FOUR Toward Empathic Innovation: Through Platform Governance Education & Research Programs: #4199990114294).

ORCID iD

Eun Jin Shin  <https://orcid.org/0000-0003-3560-0754>

Supplemental Material

Supplemental material for this article is available online.

Notes

1. As dollar stores strategically expanded into poor ethnic neighborhoods, they contributed to neighborhood stigmatization (Shannon 2021).
2. Food inspections data differentiate between different types of inspections (e.g., license, complaint, canvass).
3. With few exceptions, most dollar stores analyzed in this study opened within the same quarter when existing ones closed.

4. These crime incidents are under “small retail store” and “convenience store” categories.
5. Given that rings span across block groups, the area overlapped with each block group was calculated to estimate socioeconomic and demographic characteristics.
6. A different definition of a control ring (i.e., a 250-foot buffer of a random point in each block group with a dollar store) has been tried. However, its preopening crime level significantly differed from that of treatment rings, making it a non-ideal counterfactual.
7. Based on the restricted sample, it is confirmed that using the traditional DID approach without incorporating variables related to store closures yields results concerning the impact of dollar store openings that are comparable to baseline findings (results available upon request).
8. Six hundred eighty-three feet is the sum of the 433 and 250 feet radius of an outer and a treatment ring, respectively. See Figure A2.
9. A DID coefficient from a negative binomial regression can be interpreted as follows: for a one-unit increase in X, Y increases by $(\exp \beta - 1) \times 100\%$. Therefore, 0.17 was obtained by 0.80 (i.e., mean crime) multiplied by $(\exp (0.195) - 1)$.

References

- Askey, Amber Perenzin, Ralph Taylor, Elizabeth Groff, and Aaron Fingerhut. 2018. “Fast Food Restaurants and Convenience Stores: Using Sales Volume to Explain Crime Patterns in Seattle.” *Crime & Delinquency* 64 (14): 1836–57. <https://doi.org/10.1177/0011128717714792>.
- Bernasco, Wim. 2010. “A Sentimental Journey to Crime: Effects of Residential History on Crime Location Choice.” *Criminology; An Interdisciplinary Journal* 48 (2): 389–416. <https://doi.org/10.1111/j.1745-9125.2010.00190.x>.
- Bhattarai, Abha. 2021. “Booming business at dollar stores shows the widening gulf between haves and have-nots during pandemic.” *The Washington Post*, August 20. <https://www.washingtonpost.com/business/2021/08/20/growing-number-americans-are-relying-dollar-stores>.
- Brown, Angelica. 2018. “Community protests from local residents against the entry of dollar stores.” *KTUL*, March 4. <https://ktul.com/news/local/community-protests-new-dollar-general-says-they-want-more-grocery-stores>.
- Burkhardt, Jesse, and Chris Goemans. 2019. “The Short-Run Effects of Marijuana Dispensary Openings on Local Crime.” *The Annals of Regional Science* 63 (1): 163–89. <https://doi.org/10.1007/s00168-019-00931-0>.
- Callaway, Brantly., and Pedro H.C. Sant’Anna. 2021. “Difference-in-Differences with Multiple Time Periods.” *Journal of Econometrics* 225 (2): 200–30. <https://doi.org/10.1016/j.jeconom.2020.12.001>.
- Caspi, Caitlin Eicher, Jennifer E. Pelletier, Lisa Hamack, Darin J. Erickson, and Melissa N. Laska. 2016. “Differences in Healthy Food Supply and Stocking Practices Between Small Grocery Stores, Gas-Marts, Pharmacies and Dollar Stores.” *Public Health Nutrition* 19 (3): 540–7. <https://doi.org/10.1017/S13688980015002724>.

- Casteel, Carri, and Corinne Peek-Asa. 2000. "Effectiveness of Crime Prevention Through Environmental Design (CPTED) in Reducing Robberies." *American Journal of Preventive Medicine* 18 (4): 99–115. [https://doi.org/10.1016/S0749-3797\(00\)00146-X](https://doi.org/10.1016/S0749-3797(00)00146-X).
- Chang, Zheng, and Mi Diao. 2022. "Inter-City Transport Infrastructure and Intra-City Housing Markets: Estimating the Redistribution Effect of High-Speed Rail in Shenzhen, China." *Urban Studies* 59 (4): 870–89. <https://doi.org/10.1177/00420980211017811>.
- Chenarides, Lauren, Clare Cho, Rodolfo M. Nayga Jr, and Michael R. Thomsen. 2021. "Dollar Stores and Food Deserts." *Applied Geography* 134: 102497. <https://doi.org/10.1016/j.apgeog.2021.102497>.
- Cohen, Lawrence E., and Marcus Felson. 1979. "Social Change and Crime Rate Trends: A Routine Activity Approach." *American Sociological Review* 44 (3): 588–608. <https://doi.org/10.2307/2094589>.
- Contreras, Christopher. 2017. "A Block-Level Analysis of Medical Marijuana Dispensaries and Crime in the City of Los Angeles." *Justice Quarterly* 34 (6): 1069–95. <https://doi.org/10.1080/07418825.2016.1270346>.
- Cui, Lin, and Randall Walsh. 2015. "Foreclosure, Vacancy and Crime." *Journal of Urban Economics* 87: 72–84. <https://doi.org/10.1016/j.jue.2015.01.001>.
- D'Alessio, Stewart, and Lisa StolzenBerg. 1990. "A Crime of Convenience: The Environment and Convenience Store Robbery." *Environment and Behavior* 22 (2): 255–71. <https://doi.org/10.1177/0013916590222004>.
- Drichoutis, Andreas C., Rodolfo M. Nayga, Heather L. Rouse, and Michael R. Thomsen. 2015. "Food Environment and Childhood Obesity: The Effect of Dollar Stores." *Health Economics Review* 5 (1): 1–13. <https://doi.org/10.1186/s13561-015-0074-2>.
- Farbmacher, Helmut, Heinrich Kögel, and Martin Spindler. 2021. "Heterogeneous Effects of Poverty on Attention." *Labour Economics* 71: 102028. <https://doi.org/10.1016/j.labeco.2021.102028>.
- Feng, SQ, EL Piza, LW Kennedy, and JM Caplan. 2019. "Aggravating effects of alcohol outlet types on street robbery and aggravated assault in New York City." *Journal of Crime and Justice* 42 (3): 257–273.
- Furr-Holden, C. Debra M., Adam J. Milam, Elizabeth D. Nesoff, Renee M. Johnson, David O. Fakunle, Jacky M. Jennings, and Roland J. Thorpe Jr. 2016. "Not in My Back Yard: A Comparative Analysis of Crime Around Publicly Funded Drug Treatment Centers, Liquor Stores, Convenience Stores, and Corner Stores in One Mid-Atlantic City." *Journal of Studies on Alcohol and Drugs* 77 (1): 17–24. <https://doi.org/10.15288/jsad.2016.77.17>.
- Grigsby-Calage, Chuck, Conner C. Mullally, and Richard J. Volpe. 2021, August 1-3. *The Geography of Dollar Stores [Paper Presentation]*. Austin, TX, U.S.A: Agricultural & Applied Economics Association Annual Meeting.
- Hatten, David, and Eric L. Piza. 2022. "When Crime Moves Where Does It Go? Analyzing the Spatial Correlates of Robbery Incidents Displaced by a Place-Based Policing Intervention." *Journal of Research in Crime and Delinquency* 59 (1): 128–62. <https://doi.org/10.1177/00224278211016030>.

- He, Guojun, Yuhang Pan, and Takanao Tanaka. 2020. "The Short-Term Impacts of COVID-19 Lockdown on Urban air Pollution in China." *Nature Sustainability* 3 (12): 1005–11. <https://doi.org/10.1038/s41893-020-0581-y>.
- Heramb, Cheri. 2007. *Street and Site Plan Design Standards*. Chicago, IL: Chicago Department of Transportation. <https://www.chicago.gov/dam/city/depts/cdot/StreetandSitePlanDesignStandards407.pdf>.
- Higgs, Robert. 2020. "Cleveland Mayor Frank Jackson OKs ban on new dollar stores while city considers new regulations." *Cleveland.com*, June 25. <https://www.cleveland.com/cityhall/2020/06/cleveland-mayor-frank-jackson-oks-ban-on-new-dollar-stores-while-city-considers-new-regulations.html>.
- Howell, Debbie. 2000. "Dollar General: The Leader in the Realm of Deep Discount." *DSN Retailing Today* 39 (9): 59–62.
- Jacobs, Jane. 1961. *The Death and Life of Great American Cities*. New York: Random House.
- Jones, Megan. 2019. "Aurora man accused of attempted armed robbery at Dollar General store." *Chicago Tribune*, May 22. <https://www.chicagotribune.com/suburbs/aurora-beacon-news/ct-abn-aurora-man-armed-robbery-dollar-general-st-0523-story.html>.
- Kassab, Beth. 2013. Cities don't have to settle for ugly dollar stores. *Orlando Sentinel*, April 24. <https://www.orlandosentinel.com/opinion/os-xpm-2013-04-24-os-beth-kassab-dollar-stores-look-bad-20130424-story.html>.
- Klingseis, K. 2014. "Police interrupt drug deal in Family Dollar parking lot." *Des Moines Register*, December 3. <https://www.desmoinesregister.com/story/news/crime-and-courts/2014/12/03/police-family-dollar-drug-deal/19857035/>.
- Kubrin, Charis E., and John R. Hipp. 2016. "Do Fringe Banks Create Fringe Neighborhoods? Examining the Spatial Relationship Between Fringe Banking and Neighborhood Crime Rates." *Justice Quarterly* 33 (5): 755–84. <https://doi.org/10.1080/07418825.2014.959036>.
- Kubrin, Charis E., Gregory D. Squires, Steven M. Graves, and Graham C. Ousey. 2011. "Does Fringe Banking Exacerbate Neighborhood Crime Rates? Investigating the Social Ecology of Payday Lending." *Criminology & Public Policy* 10 (2): 437–66. <https://doi.org/10.1111/j.1745-9133.2011.00719.x>.
- MacGillis, Alec. 2020. "How dollar stores became magnets for crime and killing." *ProPublica*, June 29. <https://www.propublica.org/article/how-dollar-stores-became-magnets-for-crime-and-killing>.
- Martinez, Ramiro Jr., Richard Rosenfeld, and Dennis Mares. 2008. "Social Disorganization, Drug Market Activity, and Neighborhood Violent Crime." *Urban Affairs Review* 43 (6): 846–74. <https://doi.org/10.1177/1078087408314774>.
- Meyersohn, Nathaniel, and Moss Cohen. 2020. "Dollar General is cheap, popular and spreading across America. It's also a robbery magnet, police say." *CNN Business*, June 26. <https://edition.cnn.com/2020/06/26/business/dollar-general-robberies/index.html>.
- Miró, Fernando. 2014. "Routine Activity Theory." *The Encyclopedia of Theoretical Criminology*, 1–7. <https://doi.org/10.1002/9781118517390.wbetc198>.
- Ousey, Graham C., and Matthew R. Lee. 2002. "Examining the Conditional Nature of the Illicit Drug Market-Homicide Relationship: A Partial Test of the Theory of

- Contingent Causation.” *Criminology: An Interdisciplinary Journal* 40 (1): 73–102. <https://doi.org/10.1111/j.1745-9125.2002.tb00950.x>.
- Perkins, Craig A. 1997. *Age Patterns of Victims of Serious Violent Crime*. Washington, DC: U.S. Department of Justice.
- Perry, Andre, Jonathan Rothwell, and David Harshbarger. 2020. *Five-star Reviews, One-Star Profits: The Devaluation of Businesses in Black Communities*. Washington, DC: Brookings Institution. https://www.brookings.edu/wp-content/uploads/2020/02/2020.02_DevOfBizInBlackCommunities_Perry-Rothwell-Harshbarger-final.pdf.
- Reich, Adam. 2016. “Walmart’s Consumer Redlining.” *Contexts* 15 (4): 74–7. <https://doi.org/10.1177/1536504216685128>.
- Sampson, Robert J., and Stephen W. Raudenbush. 1999. “Systematic Social Observation of Public Spaces: A New Look at Disorder in Urban Neighborhoods.” *American Journal of Sociology* 105 (3): 603–51. <https://doi.org/10.1086/210356>.
- Sampson, Robert J., and Stephen W. Raudenbush. 2004. “Seeing Disorder: Neighborhood Stigma and the Social Construction of “Broken Windows.”” *Social Psychology Quarterly* 67 (4): 319–42. <https://doi.org/10.1177/019027250406700401>.
- Schweitzer, John H., June Woo Kim, and Juliette R. Mackin. 1999. “The Impact of the Built Environment on Crime and Fear of Crime in Urban Neighborhoods.” *Journal of Urban Technology* 6 (3): 59–73. <https://doi.org/10.1080/10630739983588>.
- Shannon, Jerry. 2021. “Dollar Stores, Retailer Redlining, and the Metropolitan Geographies of Precarious Consumption.” *Annals of the American Association of Geographers* 111 (4): 1200–18. <https://doi.org/10.1080/24694452.2020.1775544>.
- Shin, Eun Jin. 2023. “What can We Learn from Online Reviews? Examining the Reviews of Los Angeles Metro Rail Stations.” *Journal of Planning Education and Research* 43 (2): 254–67. <https://doi.org/10.1177/0739456X19870261>.
- Shrestha, Sriya. 2016. “Dollars to Dimes: Disparity, Uncertainty, and Marketing to the Poor at US Dollar Stores.” *International Journal of Cultural Studies* 19 (4): 373–90. <https://doi.org/10.1177/1367877913515869>.
- Siegel, Rachel. 2019. “As dollar stores move into cities, residents see a steep downside.” *The Washington Post*, February 15. https://www.washingtonpost.com/business/economy/as-dollar-stores-move-into-cities-residents-see-a-steep-downside/2019/02/15/b3676cbe-2f09-11e9-8ad3-9a5b113ecd3c_story.html.
- Spader, Jonathan, Jenny Schuetz, and Alvaro Cortes. 2016. “Fewer Vacants, Fewer Crimes? Impacts of Neighborhood Revitalization Policies on Crime.” *Regional Science and Urban Economics* 60: 73–84. <https://doi.org/10.1016/j.regsciurbeco.2016.07.005>.
- Stucky, Thomas D., John R. Ottensmann, and Seth B. Payton. 2012. “The Effect of Foreclosures on Crime in Indianapolis, 2003–2008.” *Social Science Quarterly* 93 (3): 602–24. <https://doi.org/10.1111/j.1540-6237.2012.00890.x>.
- Teh, Bing-Ru. 2008. *Do liquor stores increase crime and urban decay? Evidence from Los Angeles*. [Ph. D. Dissertation Paper]. University of California, Berkeley.

- Telep, Cody W., and Julie Hibdon. 2018. "Community Crime Prevention in High-Crime Areas: The Seattle Neighborhood Group Hot Spots Project." *City & Community* 17 (4): 1143–1167. <https://doi.org/10.1111/cico.12342>.
- Vargas, Tracy L. 2021. "Consumer Redlining and the Reproduction of Inequality at Dollar General." *Qualitative Sociology* 44 (2): 205–29. <https://doi.org/10.1007/s11133-020-09473-w>.
- Wager, Stefan, and Susan Athey. 2018. "Estimation and Inference of Heterogeneous Treatment Effects Using Random Forests." *Journal of the American Statistical Association* 113 (523): 1228–42. <https://doi.org/10.1080/01621459.2017.1319839>.
- Weber, Bryan, and Paolo Cappellari. 2022. "Assessing the Impact of Ferry Transit on Urban Crime." *Urban Affairs Review* 59 (6): 10780874221143047. <https://doi.org/10.1177/10780874221143047>.
- Wilson, James Q., and George L. Kelling. 1982. "Broken Windows." *Atlantic Monthly* 249 (3): 29–38.
- Wright, Richard T., and Scott H. Decker. 1997. *Armed Robbers in Action: Stickups and Street Culture*. Lebanon, NH: University Press of New England.